ENVIRONMENTAL SCIENCE & TECHNOLOGY

Does Recycling Actually Reduce Material Consumption?

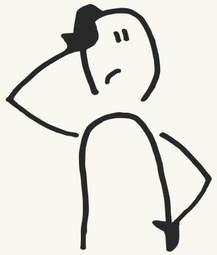


GLOBALVARMATION

TILAK
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THE CORE QUESTION

Problem



Europe is scaling up recycling —
but is it actually cutting the amount
of virgin materials we consume?

Research Questions



RQ1: Is a higher Circular Material Use (CMU) Rate
associated with lower Domestic Material Consumption (DMC)
per capita within EU countries over time?

RQ2: Does the answer change depending on how we measure
GDP (nominal vs. real) and how we classify decoupling
(trend slope vs. period endpoints)?

Three possibilities:

Recycling ↑ → Virgin material use ↓ (True decoupling)
Recycling ↑ → Total resource use flat (Partial progress)
Recycling ↑ → Total resource use ALSO ↑ (Rebound effect / Jevons paradox)

"In 2022, the EU recycled ~11.5% of its materials.
88.5% of inputs remained LINEAR." — Eurostat, 2023

FIVE OPEN DATASETS · 325 COUNTRY-YEAR OBSERVATIONS

What We Found

Source	What We Used	Years	Key Variable
Eurostat	CMU Rate	2010–2022	Recycling as % of total material use
UNEP Global MF DB	DMC & DE	2010–2022	Total material consumption & extraction
World Bank	Nominal + Real GDP	2010–2024	Economic scale & growth
OECD MF Accounts	Cross-validation of DMC/DE	2010–2022	Independent data quality check
UN SDG Database	SDG 12.2 material intensity proxy	2010–2022	Validation benchmark

Panel: 25 EU member states × 13 years = 325 observations
(Malta & Luxembourg excluded — population < 1 million)

FOUR-STAGE ANALYTICAL PIPELINE

Methodology

DATA INTEGRATION

- Downloaded & cleaned all 5 datasets
- Merged on ISO3 country code + year
- Computed per-capita indicators using World Bank population
- Added real GDP (constant 2015 USD)

DECOUPLING CLASSIFICATION

- Endpoint method: compare 2010 vs 2022 values
- Tapio Elasticity: % ΔDMC / % ΔGDP (8-category typology)
- 4-Quadrant: CMU Rate change vs DMC change direction

DESCRIPTIVE ANALYSIS

- Summary statistics across 25 countries
- Correlation matrix: CMU Rate vs DMC, DE, GDP
- CMU Rate trend lines per country (2010–2022)

PANEL REGRESSION

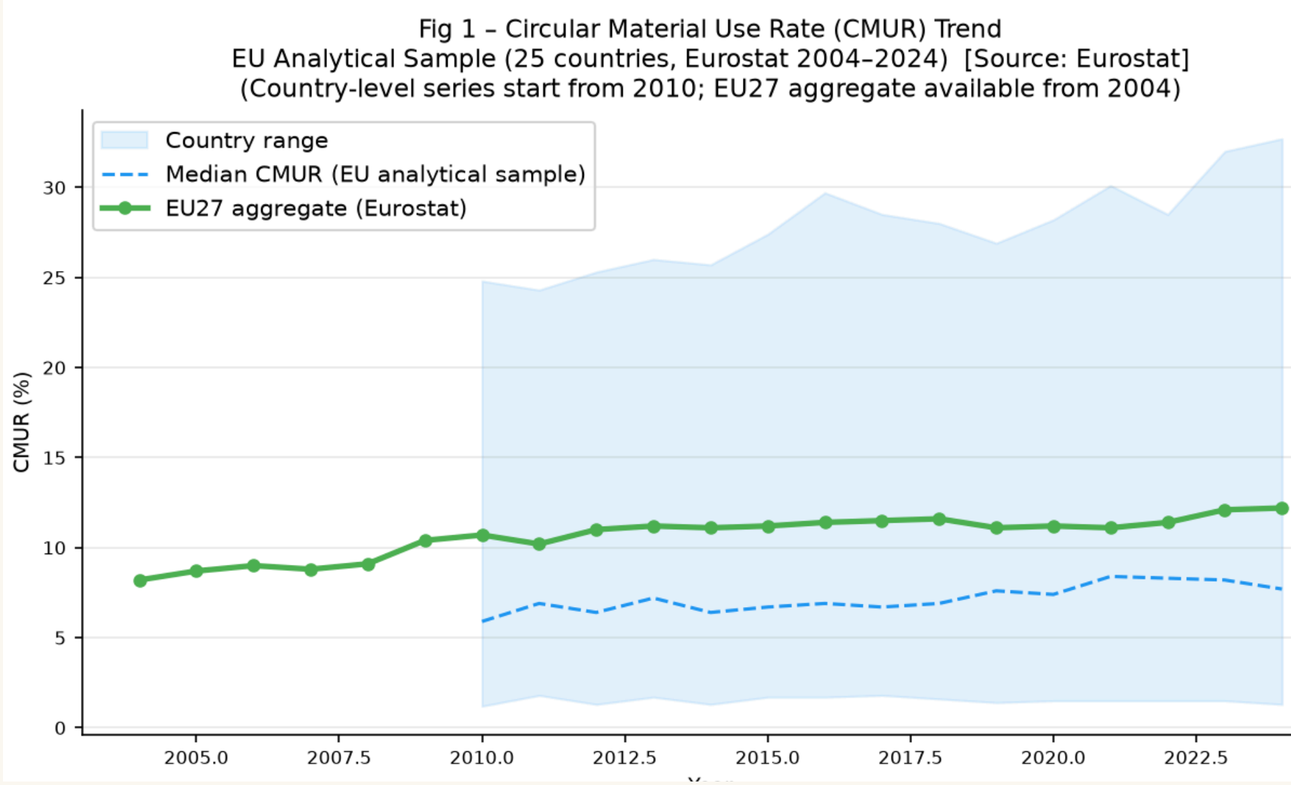
- Two-way fixed effects: country FE + year FE
- Clustered standard errors by country
- 4 two-way fixed-effects models (M1–M4) with alternative specifications

THE DATA AT A GLANCE



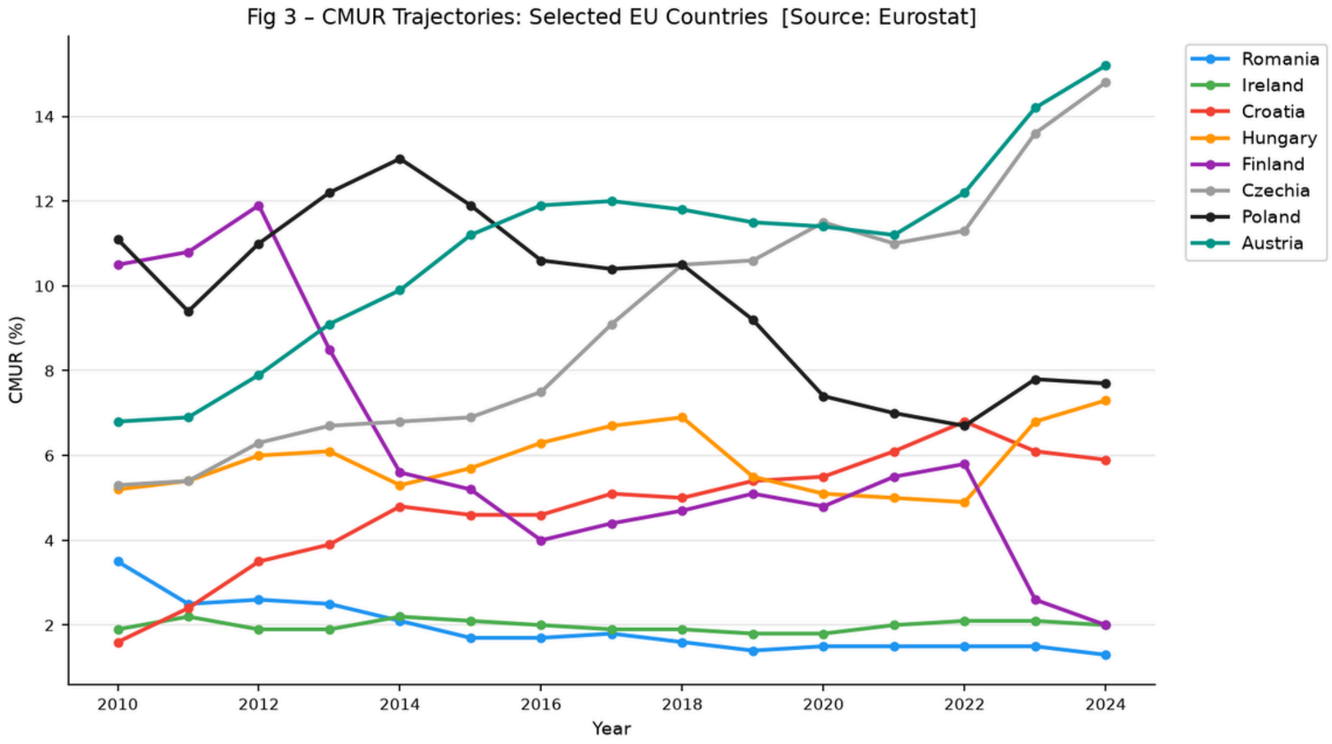
Project Objective

Indicator	Mean	Median	Min	Max
CMU Rate	8.6%	6.9%	1.2%	30.1%
DMC-total	242.2	144.7	19.6	1,297.5
Domestic Extraction	217.3	133.8	20.2	1,057.2
GDP per capita	~32,000	--	--	--

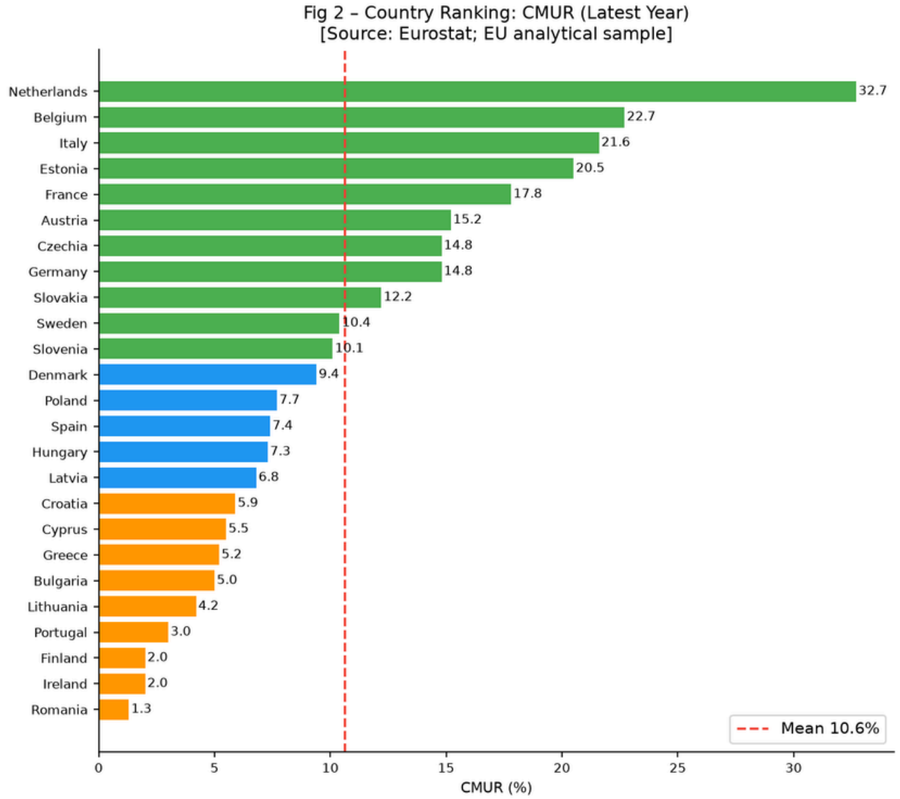


EU Aggregate CMU Rate rose from ~10.5% (2010) to ~12.5% (2022) — a 19% increase.

RECYCLING PERFORMANCE — WIDE GAP ACROSS EU MEMBERS



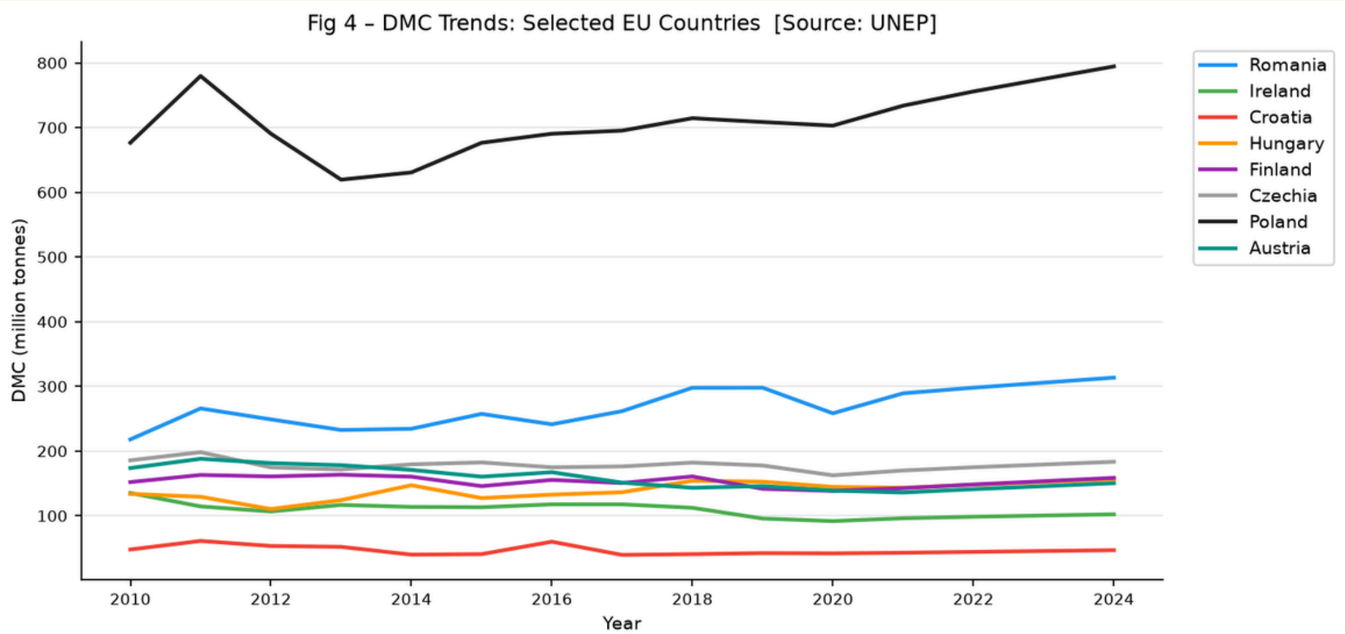
CMU Rate trajectories by country, 2010–2022

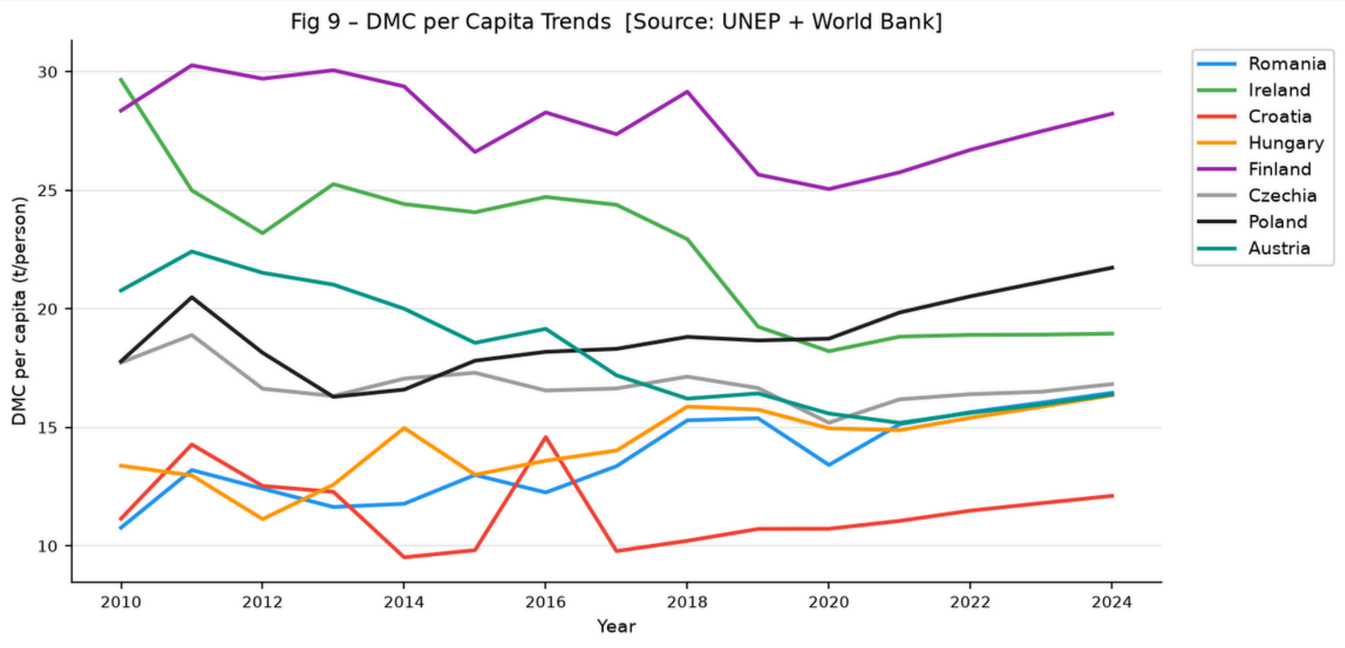


2022 country ranking — CMU Rate (%)

- Top 5 highest CMU Rate (2022): Netherlands, Belgium, France, Estonia, Italy
- Bottom 5: Romania, Ireland, Finland, Portugal and Lithuania
- Gap between best and worst: ~31 percentage points
- Eastern EU countries improving fastest from a low base

IS MATERIAL CONSUMPTION ACTUALLY FALLING?







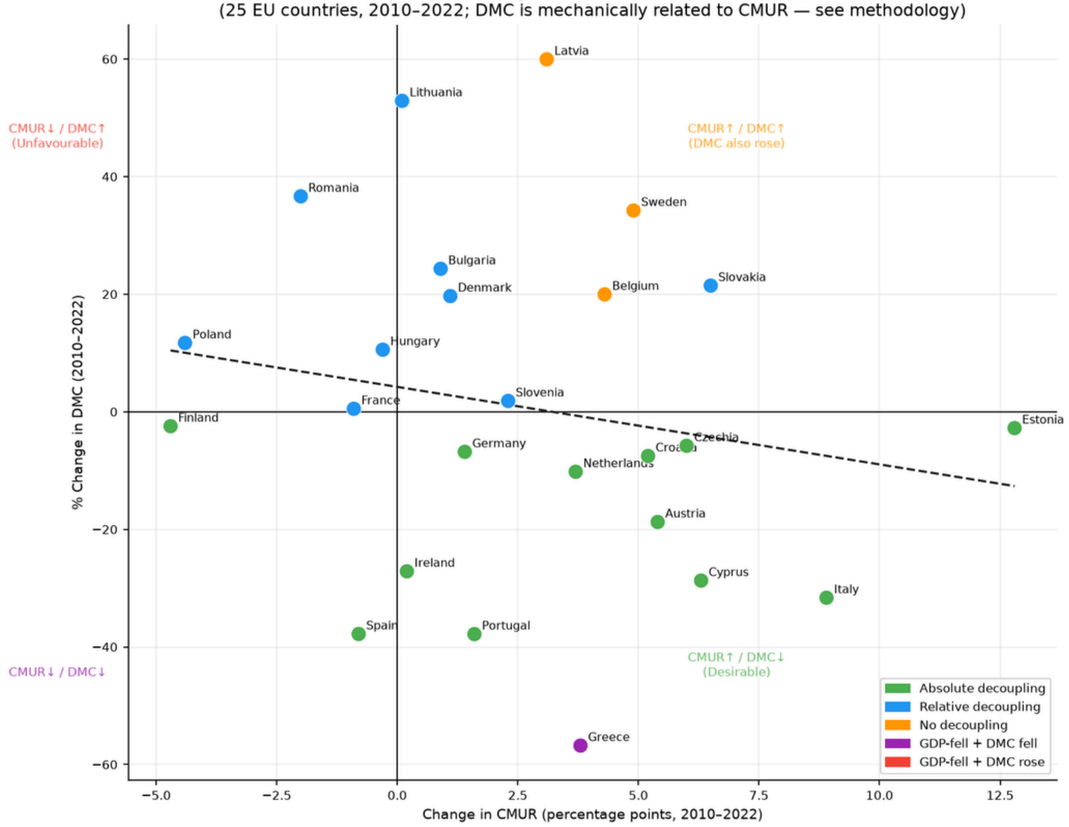
- DMC fell 2010–2013 but this was recession-driven, not policy-driven
- Post-2013 recovery: DMC increased in many countries
- Per-capita DMC remains >10 tonnes/person in Finland, Czechia and several other countries

CORRELATION ANALYSIS

DOES MORE RECYCLING = LESS VIRGIN MATERIAL USE?

Relationship	Pearson r	P-value	Interpretation
CMU Rate vs DMC per capita	-0.157	0.005	Weak negative
CMU Rate vs DE per capita	-0.19	<0.001	Weak Negative
CMU Rate vs DMC/GDP	-0.413	<0.001	DMC in both — biased
CMU Rate vs DE/GDP	-0.405	<0.001	Cleaner — no overlap

Fig 5 – Change in CMUR vs Change in DMC
(25 EU countries, 2010–2022; DMC is mechanically related to CMUR — see methodology)



CMU RATE VS DMC PER CAPITA SCATTER (2010–2022)

- **CMU Rate = Recycled Input / DMC**
- **DMC is in the denominator → mechanically creates a negative correlation**
- **This is a known indicator design flaw (Eurostat methodology note)**

THE REAL STORY — 4 QUADRANTS OF EU PERFORMANCE

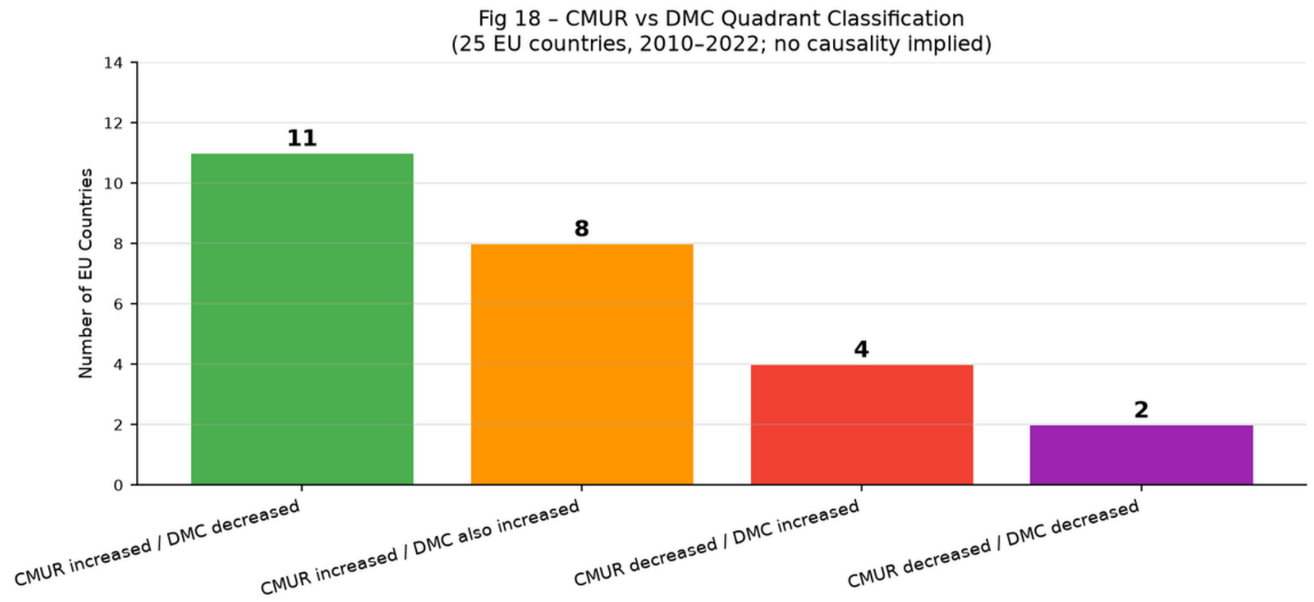
QUADRANT ANALYSIS

BEST

CMU ↑ & DMC ↓
11 countries
Austria, Germany, Ireland, Italy,
Netherlands, Portugal, Estonia,
Czech Republic, Croatia, Cyprus, Greece
→ "Recycling rising AND material use falling"

MIXED

CMU ↑ & DMC ↑
8 countries
Belgium, Sweden, Denmark, Lithuania,
Latvia, Slovakia, Slovenia, Bulgaria
→ "Recycling rising but so is total use"



RECESSION EFFECT

CMU ↓ & DMC ↓
2 countries — Spain, Finland
→ "Both fell — driven by crisis, not policy"

WORST

CMU ↓ & DMC ↑
4 countries — France, Hungary, Poland, Romania
→ "Less recycling AND more resource use"

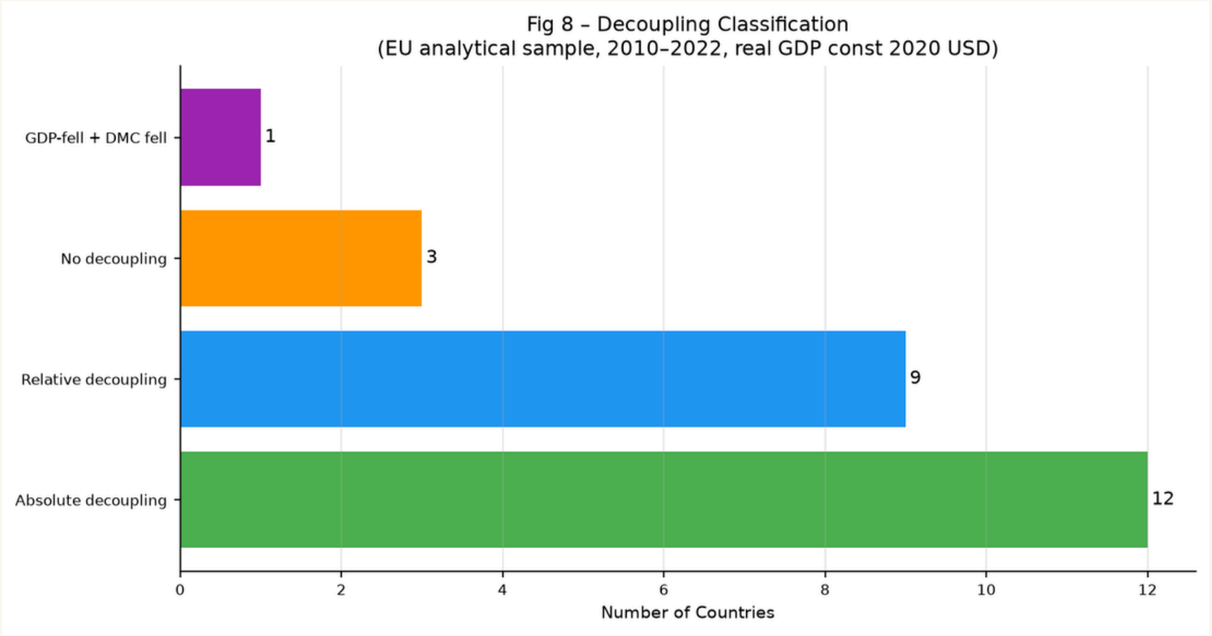
ONLY 11 OF 25 EU COUNTRIES ARE IN THE IDEAL QUADRANT. RECYCLING EXPANSION DOES NOT AUTOMATICALLY REDUCE TOTAL MATERIAL USE.

DECOUPLING CLASSIFICATION

GDP GROWTH ↗

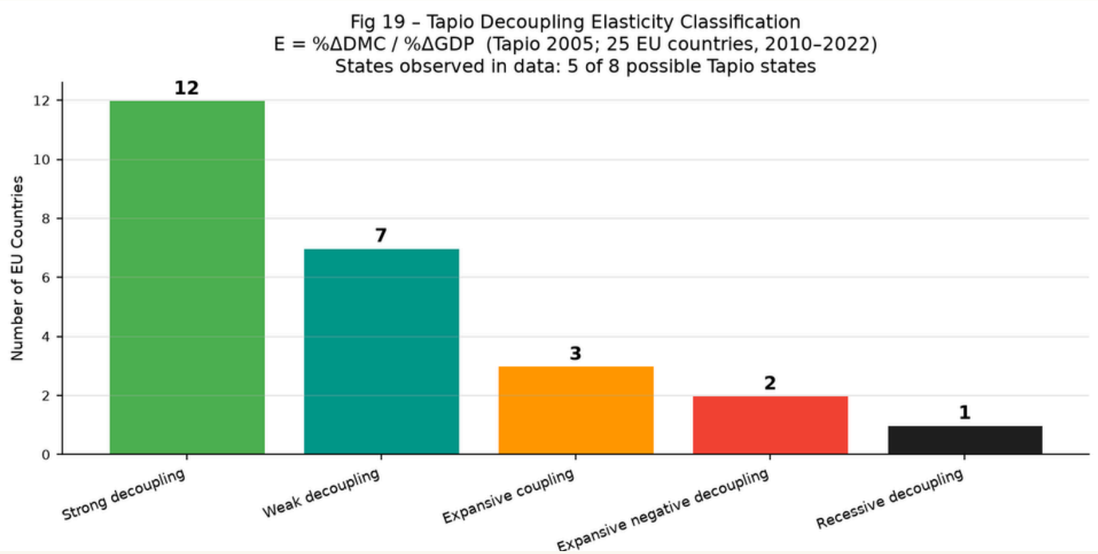


MATERIAL CONSUMPTION ↘



Category	Count	Example countries
Absolute Decoupling	12	Germany, Italy, Austria, Ireland, Netherlands
Relative Decoupling	9	France, Bulgaria, Poland, Hungary, Romania
No Decoupling	3	Belgium, Latvia, Sweden
GDP fell + DMC fell	1	Greece

12 OF 25 EU COUNTRIES ACHIEVED ABSOLUTE DECOUPLING: GDP GREW AND MATERIAL CONSUMPTION FELL (2010–2022)



TAPIO ELASTICITY CLASSIFICATION — 8-CATEGORY TYPOLOGY

GDP	DMC	E	Tapio	Directional Classificion
↑	↓	0	**Strong decoupling**	**Absolute decoupling**
↑	↑	<0.8	**Weak decoupling**	Usually **Relative decoupling**
↑	↑	0.8-1.2	**Expansive coupling**	Can be **Relative/No**
↑	↑	>1.2	**Expansive negative decoupling**	**No decoupling**

GDP MEASURE CHANGES THE DECOUPLING STORY

REAL vs NOMINAL GDP

Prior studies use NOMINAL GDP
We added REAL GDP (World Bank NY.GDP.PCAP.KD,
constant 2015 USD) as a new column

Why it matters:
Countries with high inflation or currency shifts show
inflated nominal GDP growth → appear to decouple
when they haven't structurally changed

Country	Nominal Class	Real GDP Class	Changed?
Belgium	Relative decoupling	No decoupling	YES
Italy	GDP-fell + DMC fell	Absolute decoupling	YES
Latvia	Relative decoupling	No decoupling	YES
Cyprus	Absolute decoupling	Insufficient data	YES
Greece	GDP-fell + DMC fell	insufficient data	YES

5 OF 25 COUNTRIES (20%) CHANGE CLASSIFICATION WHEN USING PRICE-ADJUSTED GDP.

ITALY'S SHIFT: EUROZONE DEBT CRISIS DEPRESSED NOMINAL GDP — NOMINAL AND REAL GDP GIVE DIFFERENT 2010–2022 GROWTH DIRECTIONS".

SLOPE vs ENDPOINT SENSITIVITY

Method A — ENDPOINT

Compare 2010 value vs 2022 value for each country
→ Simple, widely used, but sensitive to recession years

Method B — OLS TREND SLOPE

Fit regression line over all 13 years per country
→ Captures dominant long-run direction, not just endpoints

11 / 25 COUNTRIES: SAME CLASSIFICATION UNDER BOTH METHODS
14 / 25 COUNTRIES: DIFFERENT CLASSIFICATION

KEY EXAMPLE :
Ireland, Finland, France, Spain

- Endpoint method: "Absolute decoupling"
- Slope method: "DMC falling due to recession, not structural"

WHY? THESE COUNTRIES HAD MASSIVE DMC CRASH IN 2008–2013.

ENDPOINT COMPARISON FLATTERS THEM BECAUSE IT STARTS FROM A RECESSION TROUGH, NOT A NORMAL BASELINE.

Decoupling claims are sensitive to the measurement choice.
Period endpoints can be misleading when recession effects dominate the reference year.

WHAT THE EVIDENCE TELLS US

KEY FINDINGS

RECYCLING IS RISING — BUT SLOWLY

- EU CMU Rate grew from 10.5% → 12.7% (2010–2022)
- But 87.3% of materials remain linear in 2022.

RECYCLING ≠ AUTOMATIC MATERIAL REDUCTION

- 11/25 countries: CMU ↑ AND DMC ↓ (good)
- 8/25 countries: CMU ↑ AND DMC ALSO ↑ (problem)
- Jevons-type rebound in some economies

EASTERN EU: GROWTH IS OUTPACING CIRCULARITY

- Bulgaria, Lithuania, Romania: CMU ↑ but DMC ALSO ↑
- Development pressure overwhelms recycling gains

PROGRESS

WORK DONE vs REMAINING

COMPLETED

- All 5 datasets downloaded & integrated
- 325-obs panel dataset built
- CMU Rate trend visualisation
- 3-method decoupling classification
- Quadrant & Tapio analysis
- Real GDP merge (World Bank const. 2015 USD)
- Real vs nominal comparison (5 countries reclassified)
- Slope vs endpoint sensitivity (14/25 disagree)

REMAINING

- Panel regressions M1–M4 (two-way FE, clustered SE)
- CMU endogeneity: instrumental variable approach
- Municipal waste recycling rate as alternative indicator
- Material-type breakdown(metals vs biomass vs minerals)
- Trade-adjusted footprint analysis
- Final narrative report polish (Word/PDF)
- Policy recommendations section
- Literature review (12 citations)



Thank You!

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